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Stacked die transversely-excited film bulk acoustic resonator (XBAR) filters

Abstract

A stacked die XBAR filter device includes a first die containing one or more XBARs on a first surface, a second die containing one or more XBARs on a second surface, and one or more conductive vias through either the first die or the second die, where the first die is connected to the second die with the first surface facing the second surface.

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Background/Summary

RELATED APPLICATION INFORMATION
(2) This patent claims priority to co-pending U.S. provisional patent application No. 63/275,870, titled STACKED DIE XBAR FILTERS, filed Nov. 4, 2021.
BACKGROUND
Field
(3) This disclosure relates to radio frequency filters using acoustic wave resonators, and specifically to filters for use in communications equipment.
Description of the Related Art
(4) A radio frequency (RF) filter is a two-port device configured to pass some frequencies and to stop other frequencies, where “pass” means transmit with relatively low signal loss and “stop” means block or substantially attenuate. The range of frequencies passed by a filter is referred to as the “pass-band” of the filter. The range of frequencies stopped by such a filter is referred to as the

“stop-band” of the filter. A typical RF filter has at least one pass-band and at least one stop-band. Specific requirements on a passband or stop-band depend on the specific application. For example, a “pass-band” may be defined as a frequency range where the insertion loss of a filter is better than a defined value such as 1 dB, 2 dB, or 3 dB. A “stop-band” may be defined as a frequency range where the rejection of a filter is greater than a defined value such as 20 dB, 30 dB, 40 dB, or greater depending on application.

(5) RF filters are used in communications systems where information is transmitted over wireless links. For example, RF filters may be found in the RF front-ends of cellular base stations, mobile telephone and computing devices, satellite transceivers and ground stations, IoT (Internet of Things) devices, laptop computers and tablets, fixed point radio links, and other communications systems. RF filters are also used in radar and electronic and information warfare systems.

(6) RF filters typically require many design trade-offs to achieve, for each specific application, the best compromise between performance parameters such as insertion loss, rejection, isolation, power handling, linearity, size and cost. Specific design and manufacturing methods and enhancements can benefit simultaneously one or several of these requirements.

(7) Performance enhancements to the RF filters in a wireless system can have broad impact to system performance. Improvements in RF filters can be leveraged to provide system performance improvements such as larger cell size, longer battery life, higher data rates, greater network capacity, lower cost, enhanced security, higher reliability, etc. These improvements can be realized at many levels of the wireless system both separately and in combination, for example at the RF module, RF transceiver, mobile or fixed sub-system, or network levels.

(8) High performance RF filters for present communication systems commonly incorporate acoustic wave resonators including surface acoustic wave (SAW) resonators, bulk acoustic wave (BAW) resonators, film bulk acoustic wave resonators (FBAR), and other types of acoustic resonators. However, these existing technologies are not well-suited for use at the higher frequencies and bandwidths proposed for future communications networks.

(9) The desire for wider communication channel bandwidths will inevitably lead to the use of higher frequency communications bands. Radio access technology for mobile telephone networks has been standardized by the 3GPP (3.sup.rd Generation Partnership Project). Radio access technology for 5.sup.th generation mobile networks is defined in the 5G NR (new radio) standard. The 5G NR standard defines several new communications bands. Two of these new communications bands are n77, which uses the frequency range from 1300 MHz to 4200 MHz, and n79, which uses the frequency range from 4400 MHz to 5000 MHz. Both band n77 and band n79 use time-division duplexing (TDD), such that a communications device operating in band n77 and/or band n79 use the same frequencies for both uplink and downlink transmissions. Bandpass filters for bands n77 and n79 must be capable of handling the transmit power of the communications device. WiFi bands at 5 GHz and 6 GHz also require high frequency and wide bandwidth. The 5G NR standard also defines millimeter wave communication bands with frequencies between 24.25 GHz and 40 GHz.

(10) The Transversely-Excited Film Bulk Acoustic Resonator (XBAR) is an acoustic resonator structure for use in microwave filters. The XBAR is described in U.S. Pat. No. 10,491,291, titled TRANSVERSELY EXCITED FILM BULK ACOUSTIC RESONATOR. An XBAR resonator comprises an interdigital transducer (IDT) formed on a thin floating layer, or diaphragm, of a single-crystal piezoelectric material. The IDT includes a first set of parallel fingers, extending from a first busbar and a second set of parallel fingers extending from a second busbar. The first and second sets of parallel fingers are interleaved. A microwave signal applied to the IDT excites a shear primary acoustic wave in the piezoelectric diaphragm. XBAR resonators provide very high electromechanical coupling and high frequency capability. XBAR resonators may be used in a variety of RF filters including band-reject filters, band-pass filters, duplexers, and multiplexers. XBARs are well suited for use in filters for communications bands with frequencies above 3 GHz.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** includes a schematic plan view and two schematic cross-sectional views of a transversely-excited film bulk acoustic resonator (XBAR).

(2) FIG. **2** is an expanded schematic cross-sectional view of a portion of the XBAR of FIG. **1**.

(3) FIG. **3**A is an alternative schematic cross-sectional view of an XBAR.

(4) FIG. **3**B is a graphical illustration of the primary acoustic mode of interest in an XBAR.

(5) FIG. **4** is a schematic circuit diagram and layout for a high frequency band-pass filter using XBARs.

(6) FIG. **5** is a schematic circuit diagram and layout for a stacked die XBAR high frequency band-pass filter.

(7) FIGS. **6**A, **6**B, **6**C, **6**D and **6**E are a stacked die XBAR resonator filter having series resonators on one die and shunt resonators on another die.

(8) FIGS. **7**A, **7**B, **7**C, **7**D and **7**E are a stacked die XBAR resonator filter having series resonators on one die and shunt resonators on two other die.

(9) FIG. **8** is a top view of the filters of FIGS. **6**D and **7**D.

(10) FIG. **9** is a flow chart of a process for fabricating a stacked die XBAR resonator filter.

(11) FIGS. **10**A, **10**B, **10**C and **10**D are a flow chart of a process for fabricating a stacked die XBAR resonator filter.

(12) Throughout this description, elements appearing in figures are assigned three-digit or four-digit reference designators, where the two least significant digits are specific to the element and the one or two most significant digit is the figure number where the element is first introduced. An element that is not described in conjunction with a figure may be presumed to have the same characteristics and function as a previously-described element having the same reference designator or the same two least significant digits.

DETAILED DESCRIPTION

(13) Description of Apparatus

(14) The Transversely-Excited Film Bulk Acoustic Resonator (XBAR) is a new resonator structure for use in microwave filters. The XBAR is described in U.S. Pat. No. 10,491,291, titled TRANSVERSELY EXCITED FILM BULK ACOUSTIC RESONATOR, which is incorporated herein by reference in its entirety. An XBAR resonator comprises a conductor pattern having an interdigital transducer (IDT) formed on a thin floating layer or diaphragm of a piezoelectric material. The IDT has two busbars which are each attached to a set of fingers and the two sets of fingers are interleaved on the diaphragm over a cavity formed in a substrate upon which the resonator is mounted. The diaphragm spans the cavity and may include front-side and/or back-side dielectric layers. A microwave signal applied to the IDT excites a shear primary acoustic wave in the piezoelectric diaphragm, such that the acoustic energy flows substantially normal to the surfaces of the layer, which is orthogonal or transverse to the direction of the electric field generated by the IDT. XBAR resonators provide very high electromechanical coupling and high frequency capability.

(15) A piezoelectric membrane may be a part of a plate of single-crystal piezoelectric material that spans a cavity in the substrate. A piezoelectric diaphragm may be the membrane and may include the front-side and/or back-side dielectric layers. An XBAR resonator may be such a diaphragm or membrane with an interdigital transducer (IDT) formed on the diaphragm or membrane. Contact pads can be formed at selected locations over the surface of the substrate to provide electrical connections between the IDT and contact bumps to be attached to or formed on the contact pads.

(16) The resonance frequency of an XBAR is primarily set by the thickness of the diaphragm. Some broadband filters may require two different diaphragm thicknesses, which may be

implemented on two different XBAR die or chips.
(17) The following describes improved XBAR resonators, filters and fabrication techniques for stacked die XBAR resonator filters, such as accomplished using a three-dimensional packaging system that allows multiple XBAR die or chips to be stacked in a common chip or package. This stacked die XBAR may be an XBAR filter package consisting of one or more layers of stacked resonator substrates creating a 3-dimensional (3D) radio frequency (RF) structure or filter. This can be done by forming a stacked die XBAR filter device having a first die containing one or more XBARs on a first surface, a second die containing one or more XBARs on a second surface, and one or more conductive vias through either the first die or the second die, where the first die is connected to the second die with the first surface facing the second surface.
(18) Allowing resonators to be stacked vertically shrinks die footprint without reducing resonator sizes and subsequent power handling. Combining resonators from separate die allows them to be sourced or combined from different wafers with different resonator stackups (e.g., different electrode thickness and piezoelectric plate thickness) and different resonant frequencies, such as for ladder filter series and shunt resonators. The different stackups and frequencies on the vertically stacked different die lessens the range, time and expense of tuning of the resonators required to form the filter.
(19) Another approach is to use a single layer planar package where all resonators are in plane and fabricated with or on a common piezoelectric plate thickness. Thick SiO.sub.2 is deposited on some resonators (e.g., shunt resonators) to shift down their resonant frequency. These planar designs cause die size to directly scale and increase with increased resonator sizes, although a lower die size scaling is desired. Also, a thick SiO.sub.2 layer to shift down the resonant frequency has acoustic effects that adds to a native XBAR response and must be compensated for. This compensation adds frequency variation and can compromise XBAR response, which is not desirable. Also, more resonators on the piezoelectric plate require a bigger die area which is not wanted; and thick oxide can add undesired spurs to the resonator frequency response.
(20) Using thick SiO2 to shift the resonant frequency down also reduces the acoustic coupling of the resonator which is not desired. Less coupling reduces the maximum bandwidth of the assembled filter. To compensate for this and maintain bandwidth, a stronger coupling piezo material must be chosen which may have more spurious responses in-band. Compensating for the spurious may require changing IDT metal thicknesses and/or mark and pitch to sub-optimal dimensions for other filter parameters. The native frequency response varies with the piezo thickness variation across the wafer. The SiO2 deposition can also vary in thickness across the wafer adding more frequency variation and may require wafer level tuning/trimming to remove. Using the stacked die XBAR provides advantages of not only different plate thicknesses for the series and shunt die, but also allows different piezo material to be used on each of those die.
(21) FIG. **1** shows a simplified schematic top view and orthogonal cross-sectional views of a transversely-excited film bulk acoustic resonator (XBAR) **100**. XBAR resonators such as the resonator **100** may be used in a variety of RF filters including band-reject filters, band-pass filters, duplexers, and multiplexers. XBARs are particularly suited for use in filters for communications bands with frequencies above 3 GHz.
(22) The XBAR **100** is made up of a thin film conductor pattern formed on a surface of a piezoelectric plate **110** having parallel front and back surfaces **112**, **114**, respectively. The piezoelectric plate is a thin single-crystal layer of a piezoelectric material such as lithium niobate, lithium tantalate, lanthanum gallium silicate, gallium nitride, or aluminum nitride. The piezoelectric plate is cut such that the orientation of the X, Y, and Z crystalline axes with respect to the front and back surfaces is known and consistent. The piezoelectric plate may be Z-cut (which is to say the Z axis is normal to the front and back surfaces **112**, **114**), rotated Z-cut, or rotated YX cut. XBARs may be fabricated on piezoelectric plates with other crystallographic orientations.
(23) The back surface **114** of the piezoelectric plate **110** is attached to a substrate **120** that provides

mechanical support to the piezoelectric plate **110**. The substrate **120** may be, for example, silicon, sapphire, quartz, or some other material. The substrate may have layers of silicon thermal oxide (TOX) and crystalline silicon. The back surface **114** of the piezoelectric plate **110** may be bonded to the substrate **120** using a wafer bonding process, or grown on the substrate **120**, or attached to the substrate in some other manner. The piezoelectric plate is attached directly to the substrate or may be attached to the substrate via a bonding oxide layer **122**, such as a bonding oxide (BOX) layer of SiO2, or another oxide such as Al.sub.2O.sub.3.

(24) As shown in FIG. **1**, the diaphragm **115** is contiguous with the rest of the piezoelectric plate **110** around all of a perimeter **145** of the cavity **1**. In this context, “contiguous” means “continuously connected without any intervening item”. However, it is possible for a bonding oxide layer (BOX) to bond the plate **110** to the substrate **120**. The BOX layer may exist between the plate and substrate around perimeter **145** and may extend further away from the cavity than just within the perimeter itself. In the absence of a process to remove it (i.e., this invention) the BOX is everywhere between the piezoelectric plate and the substrate. The BOX is typically removed from the back of the diaphragm **115** as part of forming the cavity.

(25) The conductor pattern of the XBAR **100** includes an interdigital transducer (IDT) **130**. The IDT **130** includes a first plurality of parallel fingers, such as finger **136**, extending from a first busbar **132** and a second plurality of fingers extending from a second busbar **134**. The first and second pluralities of parallel fingers are interleaved. The interleaved fingers **136** overlap for a distance AP, commonly referred to as the “aperture” of the IDT. The center-to-center distance L between the outermost fingers of the IDT **130** is the “length” of the IDT.

(26) The first and second busbars **132**, **134** serve as the terminals or electrodes of the XBAR **100**. A radio frequency or microwave signal applied between the two busbars **132**, **134** of the IDT **130** excites a primary acoustic mode within the piezoelectric plate **110**. As will be discussed in further detail, the excited primary acoustic mode is a bulk shear mode where acoustic energy propagates along a direction substantially orthogonal to the surface of the piezoelectric plate **110**, which is also normal, or transverse, to the direction of the electric field created by the IDT fingers. Thus, the XBAR is considered a transversely-excited film bulk wave resonator.

(27) A cavity **140** is formed in the substrate **120** such that a portion **115** of the piezoelectric plate **110** containing the IDT **130** is suspended over the cavity **140** without contacting the substrate **120** or the bottom of the cavity. “Cavity” has its conventional meaning of “an empty space within a solid body.” The cavity may contain a gas, air, or a vacuum. In some case, there is also a second substrate, package or other material having a cavity (not shown) above the plate **110**, which may be a mirror image of substrate **120** and cavity **140**. The cavity above plate **110** may have an empty space depth greater than that of cavity **140**. The fingers extend over (and part of the busbars may optionally extend over) the cavity (or between the cavities). The cavity **140** may be a hole completely through the substrate **120** (as shown in Section A-A and Section B-B of FIG. **1**) or a recess in the substrate **120** (as shown subsequently in FIG. **3**A). The cavity **140** may be formed, for example, by selective etching of the substrate **120** before or after the piezoelectric plate **110** and the substrate **120** are attached. As shown in FIG. **1**, the cavity **140** has a rectangular shape with an extent greater than the aperture AP and length L of the IDT **130**. A cavity of an XBAR may have a different shape, such as a regular or irregular polygon. The cavity of an XBAR may more or fewer than four sides, which may be straight or curved.

(28) The portion **115** of the piezoelectric plate suspended over the cavity **140** will be referred to herein as the “diaphragm” (for lack of a better term) due to its physical resemblance to the diaphragm of a microphone. The diaphragm may be continuously and seamlessly connected to the rest of the piezoelectric plate **110** around all, or nearly all, of perimeter of the cavity **140**. In this context, “contiguous” means “continuously connected without any intervening item”. In some cases, a BOX layer may bond the plate **110** to the substrate **120** around the perimeter.

(29) For ease of presentation in FIG. **1**, the geometric pitch and width of the IDT fingers is greatly

exaggerated with respect to the length (dimension L) and aperture (dimension AP) of the XBAR. A typical XBAR has more than ten parallel fingers in the IDT **110**. An XBAR may have hundreds, possibly thousands, of parallel fingers in the IDT **110**. Similarly, the thickness of the fingers in the cross-sectional views is greatly exaggerated.

(30) FIG. **2** shows a detailed schematic cross-sectional view of the XBAR **100** of FIG. **1**. The cross-sectional view may be a portion of the XBAR **100** that includes fingers of the IDT. The piezoelectric plate **110** is a single-crystal layer of piezoelectrical material having a thickness ts. The ts may be, for example, 100 nm to 1500 nm. When used in filters for LTE™ bands from 3.4 GHZ to 6 GHz (e.g., bands **42**, **43**, **46**), the thickness ts may be, for example, 200 nm to 1000 nm.

(31) A front-side dielectric layer **214** may optionally be formed on the front side of the piezoelectric plate **110**. The "front side" of the XBAR is, by definition, the surface facing away from the substrate. The front-side dielectric layer **214** has a thickness tfd. The front-side dielectric layer **214** is formed between the IDT fingers **236**. Although not shown in FIG. **2**, the front side dielectric layer **214** may also be deposited over the IDT fingers **236**. A back-side dielectric layer **216** may optionally be formed on the back side of the piezoelectric plate **110**. The back-side dielectric layer may be or include the BOX layer. The back-side dielectric layer **216** has a thickness tbd. The front-side and back-side dielectric layers **214**, **216** may be a non-piezoelectric dielectric material, such as silicon dioxide or silicon nitride. The tfd and tbd may be, for example, 0 to 500 nm. tfd and tbd are typically less than the thickness is of the piezoelectric plate. The tfd and tbd are not necessarily equal, and the front-side and back-side dielectric layers **214**, **216** are not necessarily the same material. Either or both of the front-side and back-side dielectric layers **214**, **216** may be formed of multiple layers of two or more materials.

(32) The front side dielectric layer **214** may be formed over the IDTs of some (e.g., selected ones) of the XBAR devices in a filter. The front side dielectric **214** may be formed between and cover the IDT finger of some XBAR devices but not be formed on other XBAR devices. For example, a front side frequency-setting dielectric layer may be formed over the IDTs of shunt resonators to lower the resonance frequencies of the shunt resonators with respect to the resonance frequencies of series resonators, which have thinner or no front side dielectric. Some filters may include two or more different thicknesses of front side dielectric over various resonators. The resonance frequency of the resonators can be set thus "tuning" the resonator, at least in part, by selecting a thicknesses of the front side dielectric.

(33) Further, a passivation layer may be formed over the entire surface of the XBAR device **100** except for contact pads where electric connections are made to circuitry external to the XBAR device. The passivation layer is a thin dielectric layer intended to seal and protect the surfaces of the XBAR device while the XBAR device is incorporated into a package. The front side dielectric layer and/or the passivation layer may be, SiO.sub.2, Si.sub.3N.sub.4, Al.sub.2O.sub.3, some other dielectric material, or a combination of these materials.

(34) The thickness of the passivation layer may be selected to protect the piezoelectric plate and the metal conductors from water and chemical corrosion, particularly for power durability purposes. It may range from 10 to 100 nm. The passivation material may consist of multiple oxide and/or nitride coatings such as SiO2 and Si3N4 material.

(35) The IDT fingers **236** may be one or more layers of aluminum or a substantially aluminum alloy, copper or a substantially copper alloy, beryllium, tungsten, molybdenum, gold, or some other conductive material. Thin (relative to the total thickness of the conductors) layers of other metals, such as chromium or titanium, may be formed under and/or over the fingers to improve adhesion between the fingers and the piezoelectric plate **110** and/or to passivate or encapsulate the fingers. The busbars (**132**, **134** in FIG. **1**) of the IDT may be made of the same or different materials as the fingers.

(36) Dimension p is the center-to-center spacing or "pitch" of the IDT fingers, which may be referred to as the pitch of the IDT and/or the pitch of the XBAR. Dimension w is the width or

"mark" of the IDT fingers. The IDT of an XBAR differs substantially from the IDTs used in surface acoustic wave (SAW) resonators. In a SAW resonator, the pitch of the IDT is one-half of the acoustic wavelength at the resonance frequency. Additionally, the mark-to-pitch ratio of a SAW resonator IDT is typically close to 0.5 (i.e. the mark or finger width is about one-fourth of the acoustic wavelength at resonance). In an XBAR, the pitch p of the IDT is typically 2 to 20 times the width w of the fingers. In addition, the pitch p of the IDT is typically 2 to 20 times the thickness is of the piezoelectric slab **212**. The width of the IDT fingers in an XBAR is not constrained to one-fourth of the acoustic wavelength at resonance. For example, the width of XBAR IDT fingers may be 500 nm or greater, such that the IDT can be fabricated using optical lithography. The thickness tm of the IDT fingers may be from 100 nm to about equal to the width w. The thickness of the busbars (**132**, **134** in FIG. **1**) of the IDT may be the same as, or greater than, the thickness tm of the IDT fingers.
(37) FIG. **3**A is an alternative cross-sectional view of XBAR device **300** along the section plane A-A defined in FIG. **1**. In FIG. **3**A, a piezoelectric plate **310** is attached to an intermediate layer **322** of a substrate **320**. A portion of the piezoelectric plate **310** forms a diaphragm **315** spanning a cavity **340** in the substrate. The cavity **340**, does not fully penetrate the intermediate layer **322**, and is formed in the layer **322** under the portion of the piezoelectric plate **310** containing the IDT **330** of a conductor pattern (e.g., first metal or M**1** layer) of an XBAR. Fingers, such as finger **336**, of an IDT are disposed on the diaphragm **315**. Interconnection of the IDT (e.g., busbars) **330** to signal and ground paths may be through a second conductor pattern (e.g., M**2** metal layer, not shown in FIGS. **1**-**3**A) to electrical contacts on a package.
(38) Plate **310**, diaphragm **315** and fingers **336** may be plate **110**, diaphragm **115** and fingers **136** (or **236**). The cavity **340** may be formed, for example, by etching the layer **322** before attaching the piezoelectric plate **310**. Alternatively, the cavity **340** may be formed by etching the layer **322** with a selective etchant that reaches the layer **322** through one or more holes or openings **342** provided in the piezoelectric plate **310**. The diaphragm **315** may be contiguous with the rest of the piezoelectric plate **310** around a large portion of a perimeter **345** of the cavity **340**. For example, the diaphragm **315** may be contiguous with the rest of the piezoelectric plate **310** around at least 50% of the perimeter of the cavity **340**.
(39) Intermediate layer **322** may be one or more intermediate material layers attached between plate **310** and substrate **320**. An intermediary layer may be or include a bonding layer, a BOX layer, an etch stop layer, a sealing layer, an adhesive layer or layer of other material that is attached or bonded to plate **310** and substrate **320**. A layer of layers **322** may be a dielectric, an oxide, a silicon oxide, silicon nitride, an aluminum oxide, silicon dioxide or silicon nitride. Layers **322** may be one or more of any of these layers or a combination of these layers.
(40) While the cavity **340** is shown in cross-section, it should be understood that the lateral extent of the cavity is a continuous closed band area of substrate **320** that surrounds and defines the size of the cavity **340** in the direction normal to the plane of the drawing. The lateral (i.e. left-right as shown in the figure) extent of the cavity **340** is defined by the lateral edges substrate **320**. The vertical (i.e., down from plate **310** as shown in the figure) extent or depth of the cavity **340** into substrate **320**. In this case, the cavity **340** has a side cross-section rectangular, or nearly rectangular, cross section.
(41) The XBAR **300** shown in FIG. **3**A will be referred to herein as a "front-side etch" configuration since the cavity **340** is etched from the front side of the substrate **320** (before or after attaching the piezoelectric plate **310**). The XBAR **100** of FIG. **1** will be referred to herein as a "back-side etch" configuration since the cavity **140** is etched from the back side of the substrate **120** after attaching the piezoelectric plate **110**. The XBAR **300** shows one or more openings **342** in the piezoelectric plate **310** at the left and right sides of the cavity **340**. However, in some cases openings **342** in the piezoelectric plate **310** are only at the left or right side of the cavity **340**.
(42) In some cases, the substrate comprises a base substrate **320** and an intermediate layer (not

shown) to reinforce an intermediate bonding oxide (BOX) layer. Here, the first intermediate layer may be considered a part of the substrate base **320**.
(43) In some cases, layer **322** does not exist and the plate is bonded directly to the substrate **320**; and the cavity is formed in and etched into the substrate **320**.
(44) In some cases, although not shown in the figure, layer **322** is a thinner layer than the depth of the cavity such that the plate is bonded directly to layer **322**; and the cavity is formed in and etched into the layer **322** and into the substrate **320**. Here, the cavity extends completely through layer **322** and has a cavity bottom in the substrate **320**.
(45) FIG. **3**B is a graphical illustration of the primary acoustic mode of interest in an XBAR. FIG. **3**B shows a small portion of an XBAR **350** including a piezoelectric plate **310** and three interleaved IDT fingers **336**. XBAR **350** may be part of any XBAR herein. An RF voltage is applied to the interleaved fingers **336**. This voltage creates a time-varying electric field between the fingers. The direction of the electric field is primarily lateral, or parallel to the surface of the piezoelectric plate **310**, as indicated by the arrows labeled “electric field”. Due to the high dielectric constant of the piezoelectric plate, the electric field is highly concentrated in the plate relative to the air. The lateral electric field introduces shear deformation, and thus strongly excites a primary shear-mode acoustic mode, in the piezoelectric plate **310**. In this context, “shear deformation” is defined as deformation in which parallel planes in a material remain parallel and maintain a constant distance while translating relative to each other. A “shear acoustic mode” is defined as an acoustic vibration mode in a medium that results in shear deformation of the medium. The shear deformations in the XBAR **350** are represented by the curves **360**, with the adjacent small arrows providing a schematic indication of the direction and magnitude of atomic motion. The degree of atomic motion, as well as the thickness of the piezoelectric plate **310**, have been greatly exaggerated for ease of visualization. While the atomic motions are predominantly lateral (i.e. horizontal as shown in FIG. **3**B), the direction of acoustic energy flow of the excited primary shear acoustic mode is substantially orthogonal to the front and back surface of the piezoelectric plate, as indicated by the arrow **365**.
(46) An acoustic resonator based on shear acoustic wave resonances can achieve better performance than current state-of-the art film-bulk-acoustic-resonators (FBAR) and solidly-mounted-resonator bulk-acoustic-wave (SMR BAW) devices where the electric field is applied in the thickness direction. The piezoelectric coupling for shear wave XBAR resonances can be high (>20%) compared to other acoustic resonators. High piezoelectric coupling enables the design and implementation of microwave and millimeter-wave filters with appreciable bandwidth.
(47) FIG. **4** is a schematic circuit diagram and layout for a high frequency band-pass filter **400** using XBARs. The filter **400** has a conventional ladder filter architecture including three series resonators **430**A, **430**B, **430**C and two shunt resonators **440**A, **440**B. The three series resonators **430**A, **430**B, and **430**C are connected in series between a first port and a second port. In FIG. **4**, the first and second ports are labeled “In” and “Out”, respectively. However, the filter **400** is bidirectional and either port and serve as the input or output of the filter. The two shunt resonators **440**A, **440**B are connected from nodes between the series resonators to ground. All the shunt resonators and series resonators are XBARs on a single die. All or most of the resonators of FIG. **4** are XBAR resonators as noted herein.
(48) The three series resonators **430**A, B, C and the two shunt resonators **440**A, B of the filter **400** may be formed on a single plate **410** of piezoelectric material bonded to a silicon substrate (not visible). Alternatively, as will be described in further detail, the series resonators and the shunt resonators may be formed on separate plate of piezoelectric material. Each resonator includes a respective IDT (not shown), with at least the fingers of the IDT disposed over a cavity in the substrate. In this and similar contexts, the term “respective” means “relating things each to each”, which is to say with a one-to-one correspondence. In FIG. **4**, the cavities are illustrated schematically as the dashed rectangles (such as the rectangle **445**). In this example, each IDT is

disposed over a respective cavity. In other filters, the IDTs of two or more resonators may be disposed over a single cavity.

(49) FIG. **5** is a schematic circuit diagram and layout for a high frequency band-pass filter **450** using XBARs. The filter **450** has a ladder or a half ladder filter architecture including four series resonators Se**2**, Se**4**, Se**6** and Se**8** and four shunt resonators connect as parallel pairs of resonators Sh**11** and Sh**22**; Sh**31** and Sh**32**; Sh**51** and Sh**52**; and Sh**71** and Sh**72**. The four series resonators Se**2**, Se**4**, Se**6** and Se**8** are connected in series between a first port IN and a second port OUT. In this patent, the term “series” used as an adjective (e.g. series resonator, series inductor, series capacitor, series resonant circuit) means a component connected in series with other component along signal path extending from the input to the output of a network. The filter circuit **450** may be, for example, an RF transmit filter or/or an RF receive filter for incorporation into a communications device, such as a cellular telephone. The filter circuit **450** is a two-port network where one terminal of each port is typically connected to a signal ground. In FIG. **5**, the first and second ports are labeled “IN” for input and “OUT” for output, respectively. However, the filter **450** is bidirectional and either port and serve as the input or output of the filter. Each shunt resonator is connected between ground and a junction of adjacent series resonators. Shunt resonator pair Sh**11** and Sh**12** is connected from the first port to ground, although in other cases it may be connected between the second port and ground. The other three shunt resonator pairs Sh**31** and Sh**32**; Sh**51** and Sh**52**; and Sh**71** and Sh**72** are connected from nodes between the series resonators to ground. In this patent, the term “shunt” used as an adjective (e.g. shunt resonator, shunt inductor, shunt resonant circuit) means a component connected from a node along the series signal path to ground. All or most of the resonators of FIG. **5** are XBAR resonators as noted herein.

(50) Filter **450** is shown with conductive vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72**, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT that may extend through and between surfaces of die **482** and/or die **483** as shown and explained herein.

(51) The schematic diagram of FIG. **5** is simplified in that passive components, such as the inductances inherent in the conductors interconnecting the resonators, are not shown. The use of 12 acoustic wave resonators, four series resonators, and four pair of shunt resonators is exemplary. A band-pass filter circuit may include more than, or fewer than, 12 resonators; more than, or fewer than, four series resonators; and/or more or fewer than four pair of shunt resonators. For example, there may be eight series resonators and eight shunt resonators that are not in pairs.

(52) All the series resonators are XBARs on a single die **482**. A die may be a single chip, IC chip, piezoelectric plate, or substrate, such as a piece which has been diced from a processed semiconductor wafer. All the shunt resonators are XBARs on a single die **483** that is different than a die having any of the shown series resonators. In some cases, the shunt resonators are on two die, such as where the Shx**1** resonators are on one die and the Shx**2** resonators are on another die. in other cases, the series resonators may be on three or four die and the shunt resonators may be on three, four or five die.

(53) Each die may have a single plate of piezoelectric material bonded to a silicon substrate (not visible). Each resonator includes a respective IDT (not shown), with at least the fingers of the IDT disposed over a cavity in the substrate. In this and similar contexts, the term “respective” means “relating things each to each”, which is to say with a one-to-one correspondence. In this example, each IDT is disposed over a respective cavity. In other filters, the IDTs of two or more resonators may be disposed over a single cavity.

(54) FIGS. **6**A, **6**B, **6**C and **6**D are top schematic views of a stacked die XBAR resonator filter **600** having series resonators on one die **640** and shunt resonators on another die **630**. Filter **600** may be a two layer die split ladder package with two stacked die or chips that may exist in a single package.

(55) FIG. **6**A shows a package bottom or backside of filter **600** having lid surface **610** with lid backside-package contacts or contact pads **620**. The pads **620** are configured to be electrically

connected to external circuitry such as through solder bumps, or other connections (not shown) to electrically conducting signal and ground paths, wires, traces and/or the like. Each pad **620** has or is electrically connected to a conductive via that extends through at least one die of filter **600**. For filter **600**, each pad **620** has or is electrically connected to one of conductive vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72** and V**1**OUT.
(56) Conductive vias V**1**IN and V**1**OUT may be through wafer vias (TWVs) electrically connecting the one or more XBARs on a surface of one or more die of filter **600** to external circuitry, such as an input signal line and an output signal line. Conductive vias V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71** and V**72** may be TWVs electrically connecting the one or more XBARs on the second surface of a second die of filter **600** to ground, such as through or to external ground circuitry.
(57) Surface **610** may be an insulator, a dielectric such as SiO2, glass, a package material or a capping material. Surface **610** may be a silicon carrier wafer with patterned structures including contact pads **620**. It may be a printed circuit board (PCB), a high-temperature co-fired ceramics (HTCC) and/or another package for the filter **600**. It may have signal routing (e.g., vias, traces, contact pads and/or wires). In some cases, the package is a PCB laminate with copper (Cu) signal routing. It may be or include a protective encapsulating layer for the filter **600**.
(58) Pads **620**, conductor and the conductive vias herein may be gold, copper, silver, a metal, a conductive alloy or another conductor material. Although not shown, filter **600** may have gold or another conductor stud bumps attached to pads **620**. Filter **600** may then be flip-chipped bonded to a PCB or ceramic substrate, then a standard overmolding process (not shown) may be used to encapsulate the device **600**.
(59) FIG. **6**B shows a bottom die **630** of filter **600**, the die **630** having shunt resonators and conductive vias extending through surface **610** and die **630**. Die **630** may be the package lid of filter **600**. Die **630** may be a shunt resonator die/lid with a piezoelectric plate thickness=T**1** and resonance frequency=F**1**. Depending upon piezo materials, the shunt resonator thicknesses T**1** can be from 500 nm for frequencies F**1** at 3.3 GHz to 300 nm for frequencies F**1** at 6 GHz. Thus, using the stacked die XBAR provides advantages of not only different plate thicknesses for the series and shunt die, but also allows different piezo material to be used on each of those die/layers. In some cases, the frequency range of die **630** depends on the application. It could be anywhere in the RF range. The shunt resonator piezoelectric plate(s) may be greater in thickness than the series resonator plate so that the shunt resonator resonant frequency F**2** is lower than the series resonator resonant frequency F**1**.
(60) Die **630** contains or includes shunt XBARs Sh**11**, Sh**12**, Sh**31**, Sh**32**, Sh**51**, Sh**52**, Sh**71** and Sh**72**, on (e.g., formed in or on) a surface **632**. Die **630** contains conductive vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72** and V**1**OUT through surface **610** and to contacts **620**. These vias may be on or extend through: a) surface **632** and/or b) die **630** to die **640** (see FIG. **6**C).
(61) Die **630** contains conductive vias VS**2**, VS**4**, VS**6** and VS**8** on surface **632** and connected to die **640**. Conductive vias VS**2**, VS**4**, VS**6** and VS**8** may electrically connect XBARs of die **630** to XBARs or conductive vias of die **640**.
(62) Die **630** has conductive layer or traces **634** and insulator or dielectric **636** between trances **634** as shown. Traces **634** and dielectric **636** electrically connect the vias and shunt XBARs of die **630** as shown in FIGS. **5** and **6**B. Traces **634**, dielectric **636** and the vias VS**2**, VS**4**, VS**6** and VS**8** of die **630** connect the shunt XBARs of die **630** to series XBARS of die **640** as shown in FIGS. **5** and **6**E.
(63) Traces **634**, dielectric **636** and the vias V**1**IN and V**1**OUT of die **360** connect the shunt XBARs of die **630** to: a) die **640**; and b) input and output signal lines of die **640** and of surface **610** as shown in FIGS. **5-6**E. Traces **634**, dielectric **636** and the vias V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71** and V**72** of die **360** connect the shunt XBARs of die **630** to ground and to die **640** as shown in FIGS. **5-6**E.
(64) FIG. **6**C shows a top die **640** of filter **600**, the die **640** having series resonators and conductive vias extending from or through die **640**. Die **640** may be the package base or substrate of filter **600**.

Die **640** may be a series resonator die/back with a piezoelectric plate thickness=T**2** and resonance frequency=F**2**. Depending upon piezo materials, the series resonator thicknesses T**2** can be from 480 nm for frequencies F**2** at 3.1 GHz to 280 nm for frequencies F**2** at 6.5 GHz.

(65) Die **640** contains or includes series XBARs Se**2**, Se**4**, Se**6** and Se**8**, on (e.g., formed in or on) a second surface **642**. Die **640** contains conductive vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72** and V**1**OUT to surface **632** and to die **630**. Die **640** contains conductive vias VS**2**, VS**4**, VS**6** and VS**8** on surface **642** and connected to die **630**. Conductive vias VS**2**, VS**4**, VS**6** and VS**8** may electrically connect XBARs of die **640** to XBARs or conductive vias of die **630**.

(66) Die **640** has conductive layer or traces **644** and insulator or dielectric **646** between trances **644** as shown. Traces **644** and dielectric **646** electrically connect the vias and series XBARs of die **640** as shown in FIGS. **5** and **6**C.

(67) Traces **644**, dielectric **646** and the vias VS**2**, VS**4**, VS**6** and VS**8** of die **640** connect the series XBARs of die **640** to shunt XBARS of die **630** as shown in FIGS. **5**-**6**E. Traces **644**, dielectric **646** and the vias V**1**IN and V**1**OUT of die **640** connect the series XBARs of die **640** to: a) die **630**; and b) input and output signal lines of die **630** and of surface **610** as shown in FIGS. **5**-**6**E. Traces **644**, dielectric **646** and the vias V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71** and V**72** of die **640** connect the series XBARs of die **640** to ground, to die **630**, and to surface **610** as shown in FIGS. **5**-**6**E.

(68) FIGS. **6**D-**6**E shows filter **600** with surface **632** and/or vias on that surface bonded to surface **642** and/or vias on that surface.

(69) FIG. **6**D shows a top schematic view of bottom die **630** and top die **640** of filter **600**. FIG. **6**D shows the top view of the layout of shunt XBARs Sh**11**, Sh**12**, Sh**31**, Sh**32**, Sh**51**, Sh**52**, Sh**71** and Sh**72**; series XBARs Se**2**, Se**4**, Se**6** and Se**8**; conductive vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72** and V**1**OUT; and conductive vias VS**2**, VS**4**, VS**6** and VS**8**.

(70) FIG. **6**E shows a side schematic view of surface **610**, bottom die **630** and top die **640** of filter **600**. FIG. **6**E shows feature numbers **500**-**590** that will be further explained in FIGS. **10**A-**10**D and that are shown at step **580** of FIG. **10**D. FIG. **6**E shows the side view of the layout of the substrate **522**, piezoelectric plate **512** and conductor layer (e.g., IDTs) **533** of shunt XBARs Sh**11**, Sh**12**, Sh**31**, Sh**32**, Sh**51**, Sh**52**, Sh**71** and Sh**72** of die **640**; of the substrate **520**, piezoelectric plate **510** and conductor layer (e.g., IDTs) **530** of series XBARs Se**2**, Se**4**, Se**6** and Se**8** of die **630**; the bottom surface **610** and contact pads **620** (e.g., pads **590**).

(71) It shows the conductive vias VS**2**, VS**4**, VS**6** and VS**8**, such as contact pads on and between surfaces of die **630** and **640**. It shows the conductive vias V**1**IN and V**1**OUT, such as TWVs on and between surfaces of die **630** and **640**; and through die **630** to pads **620**. It shows the conductive vias V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71** and V**72**, such as contact pads on and between surfaces of die **630** and **640**; TWVs on and between surfaces of die **630** and **640**; and TWVs through die **630** to pads **620**.

(72) Each of die **630** and **640** have a substrate (not shown) having the surfaces **632** and **642** and cavities in the substrate (not shown disposed under the XBARs on the surfaces **632** and **642**. Each of die **630** and **640**, the substrate and/or surfaces **632** and **642** have a single-crystal piezoelectric plate having parallel front and back surfaces, the back surface attached to the surface of the substrate except for a portion of the piezoelectric plate forming a diaphragm that spans these cavities. Each of die **630** and **640**, the substrate and/or surfaces **632** and **642** also have an interdigital transducer (IDT) formed on the front surface of the single-crystal piezoelectric plate such that interleaved fingers of the IDT are disposed on the diaphragm, the single-crystal piezoelectric plate and the IDT configured such that a radio frequency signal applied to the IDT excites a shear primary acoustic mode within the diaphragm.

(73) Each of die **630** and **640** may be a thinned die such as by having an original thickness of the substrate polished or otherwise reduced. Each of die **630** and **640** may be or have a substrate that forms a front or lid of the filter **600**. Each of die **630** and **640** may be or have a substrate that forms a back of the filter **600**.

(74) FIGS. **6**A-**6**E may show a bonded die stack filter **600** having die **630** bonded to die **640** with the surface **642** having the shunt XBARs of die **630** facing the surface **632** having the series XBARs of die **640**. Die **640** facing die **630** means die **640** is attached to die **630** a) with the surface **642** closer to surface **632** than to the substrate of die **630** under surface **632**, and b) with the surface **632** closer to surface **642** than the substrate of die **640** under surface **632**. Facing away means the substrates are closer to each other than the others die's surface.
(75) Conductive vias VS**2**, VS**4**, VS**6** and VS**8** may be wafer bumps or contact plating extending between and electrically connecting the one or more XBARs on the first surface **632** to the one or more XBARs on the second surface **642**. Also, between die **630** and **640**, vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72** and V**1**OUT may be wafer bumps or contact plating extending between and electrically connecting the one or more XBARs, signal lines or ground lines on the first surface **632** to the one or more XBARs, signal lines or ground lines on the second surface **642**.
(76) The conductive vias VS**2**, VS**4**, VS**6** and VS**8** may be a set of contact pads, wafer bumps or contact plating on die **630** connected to, attached to, bonded to and/or touching a set of contact pads, wafer bumps or contact plating on die **640**. The conductive vias VS**2**, VS**4**, VS**6** and VS**8** may be extending between and electrically connecting the shunt XBARs on the surface **632** to the series XBARs on the surface **642**.
(77) In some cases, filter **600** is built with die **640** as the bottom or lid die with exposed surface **610** and contacts **620** on die **640** instead of on die **630**. In this case, the same concepts as shown herein apply for having the vias of the lid pads **620** extend through filter **600** to XBARs of die **630** and **640**.
(78) For example, either die **630** or **640** could be the package lid of filter **600** depending upon where via inductance of the vias of filter **600** has least impact in performance of the filter. The loss and inductance of a via can reduce shunt resonator Q, degrading filter insertion loss and rejection so typically designers would try to minimize via lengths to lower their inductance. In some cases, adding more via inductance can widen filter bandwidth slightly, which can help with process margin. Die **630** or **640** could be selected as the package lid of filter **600** based on the selected die minimizing via length and/or increasing via inductance as compared to using the other die.
(79) Stacking the resonators of die **630** and **640** instead of building a filter with these resonators unstacked (such as by being on one die, chip, wafer or plane) vertically shrinks die or package footprint without reducing resonator sizes and subsequent power handling as compared to unstacked resonators, while using the same resonators used in both filter configurations. One estimate is that stacking reduces the die footprint by 50 percent of unstacked. This reduction allows up to 2 filters **600** with the same performance to be in the footprint of one non-stacked filter.
(80) FIGS. **7**A, **7**B, **7**C, **7**D and **7**E are top schematic views of a stacked die XBAR resonator filter **700** having series resonators on one die and shunt resonators on two other die. Filter **700** may be a three layer die split ladder package with three stacked die or chips that may exist in a single package. Filter **700** may have the shunt resonators moved onto two separate layers to shrink the filter footprint or surface area from a top perspective.
(81) FIG. **7**A shows a bottom die **760** of filter **700**, the die **760** having the Shx**2** half of shunt resonators of die **630** and conductive vias extending through surface **762** and die **760**. Die **760** may be or include the package lid **710** (see FIG. **7**E) of filter **700**. Die **760** may be a shunt resonator die/lid with a piezoelectric plate thickness=T**1** and resonance frequency=F**1**. The resonators in bottom die **760** of filter **700** may have the same function as the corresponding shunt resonators in die **630** of the two layer stack of filter **600**. Die **760** may have the same piezo plate thickness and resonant frequencies as die **630**.
(82) Die **760** contains or includes shunt XBARs Sh**12**, Sh**32**, Sh**52** and Sh**72**, on (e.g., formed in or on) a surface **762**. Die **760** contains conductive vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT through surface **762** and to contacts of the lid surface. These vias may be on or extend through: a) surface **762** and/or b) die **760** to die **740** and/or **770** (see FIGS. **7**B-**7**C).

(83) Below die **760**, filter **700** may have a package bottom or backside of having a lid surface similar to surface **610** with lid backside-package contacts or contact pads similar to pads **620** for each conductive via.
(84) Die **760** contains conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** on surface **762** and connected to die **770** and die **740**. Conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may electrically connect XBARs of die **760** to XBARs or conductive vias of die **770** and die **740**.
(85) FIG. **7**B shows a middle die **770** of filter **700**, the die **770** having the Shx**1** half of shunt resonators of die **630** and conductive vias extending through surface **772** and die **770**. Die **770** may be the package mid layer of filter **700** between die **760** and **740**. Die **770** may be a shunt resonator die with a piezoelectric plate thickness=T**1** and resonance frequency=F**1**. The resonators in middle die **770** of filter **700** may have the same function as the corresponding shunt resonators in die **630** of the two layer stack of filter **600**. Die **770** may have the same piezo plate thickness and resonant frequencies as die **630**. In some cases, the resonators of die **760** and **770** are not the same as those for die **630** because shunt resonators typically have a small range of resonant frequencies and similar frequencies could be combined on each layer of die **760** and **770** to reduce deviation from the native XBAR frequency which is desired because less deviation yields fewer spurs, wider BW and higher Q.
(86) Die **770** contains or includes shunt XBARs Sh**11**, Sh**31**, Sh**51** and Sh**71**, on (e.g., formed in or on) a surface **772**. Die **770** contains conductive vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT on or through surface **772** and to vias of die **760**, die **740** and/or contacts of the lid surface. These vias may extend from or through surface **772** and/or die **770** to die **760** and/or **740** (see FIGS. **7**B-**7**C).
(87) Die **770** contains conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** on surface **772** and connected to die **760** and die **740**. Conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may electrically connect XBARs of die **770** to XBARs or conductive vias of die **760** and die **740**.
(88) FIG. **7**C shows a top die **740** of filter **700**, the die **740** having series resonators and conductive vias extending from or through die **740**. Die **740** may be the package base or substrate of filter **700**. Die **740** may be a series resonator die/back with a piezoelectric plate thickness=T**2** and resonance frequency=F**2**. The resonators in top die **740** of filter **700** may have the same function as the corresponding series resonators in die **640** of the two layer stack of filter **600**. Die **740** may have the same piezo plate thickness and resonant frequencies as die **640**.
(89) Die **740** contains or includes series XBARs Se**2**, Se**4**, Se**6** and Se**8**, on (e.g., formed in or on) a second surface **742**. Die **740** contains conductive vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT on or through surface **742** and to vias of die **760**, die **770** and/or contacts of the lid surface. These vias may extend from or through surface **742** and/or die **740** to die **760** and/or **770** (see FIGS. **7**B-**7**C).
(90) Die **740** contains conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** on surface **742** and connected to die **740** and die **760** and **770**. Conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may electrically connect XBARs of die **740** to XBARs or conductive vias of die **770** and die **760**.
(91) FIGS. **7**D-**7**E show filter **700** with surface **762** and/or vias on that surface bonded to a surface of the substrate and/or vias on the back of die **770**; and with surface **772** and/or vias on that surface bonded to surface **742** and/or vias on that surface to create ad three die stack.
(92) FIG. **7**D shows a top schematic view of bottom die **760**, middle die **770** and top die **740** of filter **700**. FIG. **7**D shows the top view of the layout of shunt XBARs Sh**11**, Sh**12**, Sh**31**, Sh**32**, Sh**51**, Sh**52**, Sh**71** and Sh**72**; series XBARs Se**2**, Se**4**, Se**6** and Se**8**; conductive vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT; and conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8**.
(93) FIG. **7**E shows a side schematic view of bottom die **760**, middle die **770** and top die **740** of filter **700**. FIG. **7**E shows feature numbers **500-590** that will be further explained in FIGS. **10**A-**10**D and that are shown at step **580** of FIG. **10**D. FIG. **7**E shows the side view of the layout of the

substrate **522**′, piezoelectric plate **512**′ and conductor layer (e.g., IDTs) **533**′ of shunt XBARs Sh**11**, Sh**31**, Sh**51** and Sh**71** of die **760**; of the substrate **522**, piezoelectric plate **512** and conductor layer (e.g., IDTs) **533** of shunt XBARs Sh**11**, Sh**31**, Sh**51** and Sh**71** of die **770**; of the substrate **520**, piezoelectric plate **510** and conductor layer (e.g., IDTs) **530** of series XBARs Se**2**, Se**4**, Se**6** and Se**8** of die **740**; and the bottom surface **710** and contact pads **720** (e.g., pads **590**′).

(94) It shows the conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8**, such as contact pads on and between surfaces of die **760**, **770** and **740**. It shows the conductive vias V**2**IN and V**2**OUT, such as TWVs on and between surfaces of die **760**, **770** and **740**; and through die **770** and **760** to pads **720**. It shows the conductive vias V**21**, V**23**, V**25** and V**27**, such as contact pads on and between surfaces of die **760**, **770** and **740**; TWVs on and between surfaces of die **760** and **770**; and TWVs through die **760** and **770** to pads **720**.

(95) FIG. **7**A-**7**E may show a bonded die stack filter **700** having die **760** bonded to die **770** which is bonded to die **740**, with the surface **762** having the shunt XBARs of die **760** facing the surface **772** having the shunt XBARs of die **770**; and the surface **772** having the shunt XBARs of die **770** facing away from the surface **742** having the series XBARs of die **740**.

(96) Conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may be wafer bumps or contact plating extending between and electrically connecting the one or more XBARs on the surface **742** to the one or more XBARs on the surface **772**. Also, between die **770** and **740**, vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT may be wafer bumps or contact plating extending between and electrically connecting the one or more XBARs, signal lines or ground lines on the surface **772** to the one or more XBARs, signal lines or ground lines on the second surface **742**. The conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may be a set of contact pads, wafer bumps or contact plating on die **770** connected to, attached to, bonded to and/or touching a set of contact pads, wafer bumps or contact plating on die **740**. The conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may be extending between and electrically connecting the shunt XBARs on the surface **772** to the series XBARs on the surface **742**.

(97) Between die **770** and **760**, vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT may be TWVs extending through surface **772** and die **770** and electrically connecting the one or more XBARs, signal lines or ground lines on the surface **772** to the one or more XBARs, signal lines or ground lines on the second surface **762**. These vias may also include a set of contact pads, wafer bumps or contact plating on back or substrate of die **770** connected to, attached to, bonded to and/or touching a set of contact pads, wafer bumps or contact plating on die **760**.

(98) The conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may be a set of TWVs through die **770** connected to, attached to, bonded to and/or touching a set of contact pads, wafer bumps or contact plating on die **760**. The conductive vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** may be extending between and electrically connecting the shunt XBARs on the surface **772** to the shunt XBARs on the surface **762**.

(99) Vias V**2**S**2**-V**2**S**8** of FIGS. **7**A-**7**E may function the same as vias VS**2**-VS**8** of FIGS. **6**A-D. Vias V**2***x* of FIGS. **7**A-**7**E may function the same as the combination of vias Vx**1** and Vx**2** of FIG. **6**A-D. This functionality may also correspond to the via connection mapping in FIG. **5**.

(100) Each of die **760**, **770** and **740** have a substrate (not shown) having the surfaces **762**, **772** and **742** and cavities in the substrate (not shown disposed under the XBARs on the surfaces **762**, **772** and **742**. Each of die **760**, **770** and **740**, the substrate and/or surfaces **762**, **772** and **742** have a single-crystal piezoelectric plate having parallel front and back surfaces, the back surface attached to the surface of the substrate except for a portion of the piezoelectric plate forming a diaphragm that spans these cavities. Each of die **760**, **770** and **740**, the substrate and/or surfaces **762**, **772** and **742** also have an interdigital transducer (IDT) formed on the front surface of the single-crystal piezoelectric plate such that interleaved fingers of the IDT are disposed on the diaphragm, the single-crystal piezoelectric plate and the IDT configured such that a radio frequency signal applied to the IDT excites a shear primary acoustic mode within the diaphragm.

(101) Each of die **760**, **770** and/or **740** may be a thinned die such as by having an original thickness of the substrate polished or otherwise reduced. Each of die **760** or **740** may be or have a substrate that forms a front or lid of the filter **700**.
(102) Die **760**, **770** and **740** have conductive layer or traces **764**, **774** and **744** and insulator or dielectric **766**, **776** and **746** between the trances as shown. These traces and dielectric electrically connect the vias and XBARs of these die as shown in FIGS. **5** and **7**A-**7**E. These traces, dielectric and the vias V**2**S**2**, V**2**S**4**, V**2**S**6** and V**2**S**8** connect the shunt XBARs of die **760** to the shunt XBARs of die **770** and to the series XBARS of die **740** as shown in FIGS. **5** and **7**A-**7**E. These traces, dielectric and the vias V**1**IN and V**1**OUT connect the shunt XBARs of die **760** to: a) the shunt XBARs of die **770**; b) the series XBARS of die **740**; and c) the input and output signal lines of filter **700** as shown in FIGS. **5** and **7**A-**7**E. These traces, dielectric and the vias V**21**, V**23**, V**25** and V**27** connect the shunt XBARs of die **760** and **770** to ground and to each other as shown in FIGS. **5** and **7**A-**7**E.
(103) In some cases, filter **700** is built with die **740** as the bottom or lid die with an exposed surface and contacts instead of on die **760** as the lid die. In this case, the same concepts as shown herein apply for having the vias of the lid pads extend through filter **700** to XBARs of die **760**, **770** and **740**. For example, either die **760** or **740** could be the package lid of filter **700** depending upon where via inductance of the vias of filter **700** has least impact in performance of the filter. The loss and inductance of a via can reduce shunt resonator Q, degrading filter insertion loss and rejection so typically designers would try to minimize via lengths to lower their inductance. In some cases, adding more via inductance can widen filter bandwidth slightly, which can help with process margin. Die **760** or **740** could be selected as the package lid of filter **700** based on the selected die minimizing via length and/or increasing via inductance as compared to using the other die.
(104) FIG. **8** is a top view **800** of the filters of FIGS. **6** and **7**. View **800** shows FIG. **7**D to the left of FIG. **6**D in order to give a three-layer versus two-layer split ladder footprint size comparison. View **800** shows that the three-layer package is 48% the area of the two layer package, such as where the filter **700** has a footprint of 713×876 um and the filter **600** has a footprint of 1000×1300 um. The smaller footprint of the three-layer package is more desirable because it vertically shrinks die footprint without reducing resonator sizes and subsequent power handling as compared to the two layer stacked resonators, while using the same resonators used in both filter configurations. This reduction allows up to 2 filters **700** with the same performance to be in the footprint of one filter **600**.
(105) Stacking the resonators of die **760**, **770** and **740** instead of building a filter with these resonators unstacked (such as by being on one die, chip, wafer or plane) vertically shrinks die or package footprint without reducing resonator sizes and subsequent power handling as compared to unstacked resonators, while using the same resonators used in both filter configurations. One estimate is that stacking reduces the die footprint by 75 percent of unstacked. This reduction allows up to 4 filters **700** to be in the footprint of one non-stacked filter with the same performance.
(106) Description of Methods
(107) FIG. **9** is a flow chart of a process **900** for fabricating a stacked die XBAR resonator filter. Process **900** may be fabricating an acoustic resonator device having a plurality of die. The process **900** includes a package assembly flow for an RF bandpass filter. The process **900** may be the forming or may be included in the forming of filter **450**, **600** or filter **700**. The flow chart of FIG. **9** includes only major process steps. Various conventional process steps (e.g. surface preparation, chemical mechanical processing (CMP), cleaning, inspection, deposition, photolithography, baking, annealing, monitoring, testing, etc.) may be performed before, between, after, and during the steps shown in FIG. **9**. The process **900** starts at **910** with wafer **1** containing a plurality of a first XBAR die with a first stackup, and ends after dicing a package at **970** to form a number of completed XBAR RF bandpass filters. For example, the first XBAR die may be the XBAR die **630** and the completed RF bandpass filters may be the filter **600**.

(108) At **910**, semiconductor processing to form wafer**1** XBAR stackup **1** is performed to form a number of XBAR filter die, such as die **640** as noted herein at locations on a wafer. Step **910** may include forming contact pads on the front surface of the die for vias, such as vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72**, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT. In some cases, only pads for vias V**1**IN, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT are formed.
(109) At **920**, semiconductor processing to form wafer**2** XBAR stackup **2** is performed to form a number of XBAR filter die, such as die **630** as noted herein at locations on a wafer. Step **920** may include forming contact pads on the front surface of the die for vias, such as vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72**, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT. In some cases, only pads for vias V**1**IN, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT are formed on the front surface. Step **920** may include forming TWVs on the back surface of the die for vias, such as vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72**, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT that extend through the die surface, such as surface **610**.
(110) Steps **910** and **920** may be XBAR wafer die fabrications to form XBARS described at FIGS. **1-3** that are formed on the surfaces of the die, such as die **640** and **630**. For example, steps **910** and **920** include forming IDTs with interleaved fingers on diaphragms formed over cavities of the substrates.
(111) At **930** and **940**, the two die of steps **910** and **920** (such as die **640** and die **630**) are RF probed and tuned, such as by separately probing, measuring the admittance of and tuning the admittance of each XBAR of the die. Tuning may include adding, thinning or thickening frontside and/or backside dielectric formed on the piezoelectric plate of the XBAR.
(112) At **950**, the second of the two die, such as die **630**, have their piezoelectric plates etched; substrates etched or drilled to form thru silicon vias through the die; and the vias filled with conductor in the die. Step **950** may be or include forming conductive vias, such as vias V**1**IN, V**11**, V**12**, V**31**, V**32**, V**51**, V**52**, V**71**, V**72**, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT. In some cases, only vias V**1**IN, VS**2**, VS**4**, VS**6**, VS**8** and V**1**OUT are formed. Etching the piezoelectric plate to form the vias may be performed separately from and in addition to other etches of that plate.
(113) In other cases, step **950** may be performed on the first of the two die, such as die **640**, to form the vias through that die, as discussed herein.
(114) At **960**, wafer bumps are formed on contact pads vias of the first die, such as die **630**, of that wafer; and the front surfaces having the contact pads facing each other of the two wafers having the two die (such as die **630** and **640**) are ultrasonic bonded together. Step **960** may be attaching front surfaces of die **630** to front surfaces of die **640**. In other cases, the bumps are formed on the second die, such as die **640**. Bumps may not be formed when other connection or attachment of the contact pads is used, such as by direct contact and heating or other conductive bonding of the pads of the die to each other. Bonding other than ultrasonic may be performed in place of or in addition to the ultrasonic bonding. Step **960** may be or include hard lid bonding. Current hard lid bonding process may use thermocompression bonding with ultrasonics.
(115) Step **960** forms ladder filters, such as filters **600**, at the locations of the two bonded die on the wafers. Step **960** may include forming a lid surface and contacts, such as surface **610** and contacts **620**. Step **960** electrically connects the conductive vias the XBARs of the first die (such as die **630**) to XBARs of the second die (such as die **640**) as noted herein.
(116) The steps **910**, **920**, **950** and **960** may include forming wafer bumps or contact plating extending between and electrically connecting the XBARs of the first and second. These steps may include forming TWVs electrically connecting the XBARs of the first die to ground contact pads and/or circuitry; and forming TWVs electrically connecting the XBARs of the first and second die to input/output signal contact pads and/or circuitry.
(117) At **970**, the bonded wafers having the first and second die are diced to separate the ladder filters, such as filters **600**. Step **970** may dice the locations on the bonded die wafer that form the filters to separate each of the filters from the others. Additional processing may be performed at

step **970** to create chips or packages each having one or more of the filters. Step **970** may be dicing a wafer package to form a number of completed XBAR RF band pass ladder filters, such as filters **600**.
(118) Process **900** may be used to form three-layer filters such as the filter **700** instead of two-layer filters such as the filter **600**, by including another instance or leg of steps **920**, **940** and **950** to form die **770** on one wafer, while the first instance of these three steps forms die **760** on another wafer, both with the vias noted for die **760** and **770**. Then at step **960**, all three wafers are bonded to form filters **700** at locations on the bonded wafer; and at step **970** the filter **700** are diced from the bonded wafer. Forming die **760** and **770** may include forming conductive vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT of and through die **760** as noted herein; and conductive vias V**2**IN, V**21**, V**23**, V**25**, V**27**, V**2**S**2**, V**2**S**4**, V**2**S**6**, V**2**S**8** and V**2**OUT of and through die **770** as noted herein. In this case, the step **950** and **960** may include forming wafer bumps or contact plating extending between and electrically connecting the XBARs of die **760**, **770** and **740**. Here, steps **960** may include attaching a front surface of die **760** to a back surface of die **770** and attaching a front surface of die **770** to a front surface of die **740**.
(119) FIGS. **10**A, **10**B, **10**C and **10**D are a flow chart of a process **1000** for fabricating a stacked die XBAR resonator filter. Process **1000** may be fabricating an acoustic resonator device having a plurality of die. The process **1000** includes a package assembly flow for an exemplary RF bandpass filter. The process **1000** may be the forming or may be included in the forming of filter **450**, **600** or filter **700**. The flow chart of FIG. **10** includes only major process steps. Various conventional process steps (e.g. surface preparation, chemical mechanical processing (CMP), cleaning, inspection, deposition, photolithography, baking, annealing, monitoring, testing, etc.) may be performed before, between, after, and during the steps shown in FIG. **10**. The process **1000** starts at **515** and ends after dicing a package at **580** to form a number of completed XBAR RF band pass filters **600**.
(120) At **530**, wafer**1** LN**1** M**1** are performed to form piezoelectric plate **510** on substrate **520** and then to form conductor layer having IDT **530** on the plate **510** to form device **501**. Step **530** may include or be part of step **910**.
(121) At **535**, wafer**2** LN**2** M**2** are performed to form piezoelectric plate **511** on substrate **521** and then to form conductor layer having IDT **531** on the plate **511** to form device **502**. The structures of step **535** are on a different die, wafer and/or substrate than those of step **530**. Step **530** may include or be part of step **920**.
(122) Steps **530** and **535** may be respective XBAR piezoelectric plate **510** and **511** and electrode layer **530** and **531** package processing on silicon base wafers **520** and **521** to form XBARS described at FIGS. **1-3** on the surfaces of die **640** and **630**, respectively. For example, steps **530** and **535** include forming IDTs with interleaved fingers on diaphragms formed over cavities of the substrates.
(123) At **540**, thru wafer vias (TWV) **560** are formed on device **502** and filled with conductive material **550** such as metal to form device **503**. Step **540** also includes etching and/or other processing of plate **511** and layer **531** to form openings **550** above the TWVs and thus form plate **512** and layer **532**. Step **540** may also include etching and/or other processing of plate **511** and layer **531** to form the XBARs of die **630**.
(124) At **545**, via plate-up is performed on device **503** to fill openings **550** with conductive material **551** such as metal, thus forming layer **533** over the layer **512** and forming device **504**. Step **545** may include electroplating to form the material **551**. It may also include polishing layer **533**. Steps **540** and **545** may include or be part of step **920** and/or **950**.
(125) At **550**, wafer bump/contact plating is performed on device **501** to form solder bumps and/or contact pads **570** of conductive material such as metal on layer **530**, thus forming device **505**. Step **550** may include patterning and etching to form contact pads **570**. Device **505** may be die **640**.
(126) Step **550** also includes wafer bump/contact plating performed on device **504** to form solder

bumps and/or contact pads **580** of conductive material such as metal on layer **533**, thus forming device **506**. Step **550** may include patterning and etching to form contact pads **580**. Device **506** may be die **630**. Step **550** may include or be part of step **910**, **920** and/or **950**.

(127) At **560**, wafer ultrasonic bonding is performed on devices **505** and **506** to bond those devices together, thus forming device **507**. Step **560** may include flipping device **505** or **506** and aligning the solder bumps and/or contact pads **570** with pads **580**. Step **560** may include flip-chip bonding the solder bumps and/or contact pads **570** to pads **580**. Step **560** may include or be part of bonding of step **960**. Device **507** may be part of device **600**.

(128) At **565**, lid thinning is performed on device **507** to reduce the thickness of substrate **521**, thus forming substrate **522** and device **508**. Step **565** may include polishing the bottom surface of substrate **521** at least to the conductor **560** so that the conductor is exposed at the bottom surface. The bottom surface of substrate **522** may be surface **610**. Step **565** may include or be part of bonding of step **920** and/or **950**.

(129) At **570**, contact plating is performed on device **508** to form contact plates **590** on the bottom surface of substrate **522**, thus forming device **509**. Plates **590** may be formed on and attached to conductor **560**. Step **570** may include patterning and etching to form contact plates **590**. Plates **590** may be pads **620**. Device **509** may be device **600** or a bonded wafer having devices **600** prior to dicing as noted at process **900**. Step **570** may include or be part of step **920** and/or **950**.

(130) At **580**, die singulation is performed on device **509** to separate ones of filter **600** from other ones of filter **600**, thus forming device **510**. Step **580** may include wafer dicing a bonded wafer having devices **600** as noted at step **970** of process **900**. Device **510** is device **600**.

(131) Process **1000** may be used to form filters **700** instead of **600** by including another instance or set of steps **535**, **540**, **545** and **550** to form die **770** on one wafer, while the first instance of these six steps form die **760** on another wafer, both with the vias noted for die **760** and **770**. Then at a first of steps **560** and **565**, die **770** is bonded to die **740**; and die **770** is thinned. Then at another of steps **560** and **565** die **776** is bonded to die **770** and **740**; and die **760** is thinned. Then, at steps **570** and **580** the contact plates **720** are formed and the filters **700** are diced from the bonded wafer.

(132) Although the description herein relate to an XBAR filter, the same concepts can be applied to a filter that each of one, some or all of the XBARs with a surface acoustic wave resonator (SAW), a bulk acoustic wave (BAW) resonator, a film bulk acoustic wave (FBAW) resonator, a temperature compensated surface acoustic wave resonator (TC-SAW), or a solidly-mounted transversely-excited film bulk acoustic resonator (SM-XBAR). They could also be any of a number of combinations of these types of resonators.

(133) Also, although some embodiments described herein are for filters having 4 series and 8 shunt resonators, the concepts described can be applied to filters having one shunt resonator instead of the pairs, having fewer or more than 4 series resonators and having fewer or more than 4 shunt resonator pair (or single shunt resonators). For example, the concept can be applied to filter **400**.

Closing Comments

(134) Throughout this description, the embodiments and examples shown should be considered as exemplars, rather than limitations on the apparatus and procedures disclosed or claimed. Although many of the examples presented herein involve specific combinations of method acts or system elements, it should be understood that those acts and those elements may be combined in other ways to accomplish the same objectives. With regard to flowcharts, additional and fewer steps may be taken, and the steps as shown may be combined or further refined to achieve the methods described herein. Acts, elements and features discussed only in connection with one embodiment are not intended to be excluded from a similar role in other embodiments.

(135) As used herein, “plurality” means two or more. As used herein, a “set” of items may include one or more of such items. As used herein, whether in the written description or the claims, the terms “comprising”, “including”, “carrying”, “having”, “containing”, “involving”, and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional

phrases “consisting of” and “consisting essentially of”, respectively, are closed or semi-closed transitional phrases with respect to claims. Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. As used herein, “and/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items.

Claims

1. A stacked die XBAR filter device comprising: a first die containing one or more shunt XBARs on a first surface, the one or more shunt XBARs having a first piezoelectric layer with a first thickness T**1** and a first interdigital transducer having interleaved fingers on the first piezoelectric layer; a second die containing one or more series XBARs on a second surface, the one or more series XBARs having a second piezoelectric layer with a second thickness T**2** that is less than the first thickness T**1** and a second interdigital transducer having interleaved fingers on the second piezoelectric layer; and one or more conductive vias through either the first die or the second die, wherein the first die is connected to the second die with the first surface facing the second surface, and wherein the interleaved fingers of the first interdigital transducer have a different thickness than the interleaved fingers of the second interdigital transducer.
2. The filter device of claim 1, wherein the one or more conductive vias include wafer bumps or contact plating extending between and electrically connecting the one or more shunt XBARs on the first surface to the one or more series XBARs on the second surface.
3. The filter device of claim 2, wherein the one or more conductive vias include: first through wafer vias (TWVs) for electrically connecting at least some of the one or more series XBARs to input and output signal external circuitry, and second TWVs for electrically connecting the one or more shunt XBARs to ground.
4. The filter device of claim 1, wherein: a first conductive via of the one or more conductive vias electrically connects the one or more shunt XBARs to the one or more series XBARS, a second conductive via of the one or more conductive vias electrically connects the one or more shunt XBARs to ground, and a third conductive via of the one or more conductive vias electrically connects at least some of the one or more series XBARs to external circuitry.
5. The filter device of claim 1, wherein: the first die further comprises a first substrate having the first surface and first cavities in an intermediate layer of the first substrate that are disposed under the one or more shunt XBARS, the second die further comprises a second substrate having the second surface and second cavities in an intermediate layer of the second substrate that are disposed under the one or more series XBARs, and the one or more conductive vias are through either the first substrate or the second substrate.
6. The filter device of claim 5, wherein the one or more shunt XBARs and the one or more series XBARs each include the respective piezoelectric layer having a portion that forms a diaphragm that is over the first cavities or the second cavities, wherein the respective interdigital transducer is on a surface of the respective piezoelectric layer and the interleaved fingers are on the respective diaphragm, and wherein the piezoelectric layer and the interdigital transducer are configured such that a radio frequency signal applied to the interdigital transducer excites a primarily shear acoustic mode within the respective diaphragm.
7. The filter device of claim 5, wherein: the conductive vias are through the second die, the second substrate is thinned, the second die forms a lid of the filter device, the first substrate forms a back of the filter device.
8. The filter device of claim 1, further comprising: a third die containing one or more XBARs on a

third surface; and one or more conductive vias through the second die or the third die for electrically connecting the second die to the third die, wherein the second die is connected to the third die with the second surface facing away from the third surface.

9. A stacked die XBAR filter device, comprising: a first piezoelectric layer having a first thickness T**1** between first front and back surfaces, the first back surface attached to a surface of a first substrate, portions of the first piezoelectric layer forming one or more first diaphragms that are over respective cavities in an intermediate layer of the first substrate; a first conductor pattern on the first front surface, the first conductor pattern including a first plurality of interdigital transducers (IDTs) of a respective first plurality of acoustic resonators, first interleaved fingers of each of the plurality of first IDTs disposed on one of the one or more first diaphragms; a second piezoelectric layer having a second thickness T**2** between second front and back surfaces, the second back surface attached to a surface of a second substrate, portions of the second piezoelectric layer forming one or more second diaphragms that are over respective cavities in an intermediate layer of the second substrate; a second conductor pattern on the second front surface, the second conductor pattern including a second plurality of interdigital transducers (IDTs) of a respective second plurality of acoustic resonators, second interleaved fingers of each of the plurality of second IDTs disposed on one of the one or more second diaphragms; and one or more conductive vias through the first substrate, wherein the first conductor pattern is connected to the second conductor pattern with the first front surface facing the second front surface, wherein the second thickness T**2** of the second piezoelectric layer is different than the first thickness T**1** of the first piezoelectric layer, and wherein the first interleaved fingers of each of the plurality of first IDTs has a different thickness than the second interleaved fingers of each of the plurality of second IDTs.

10. The filter device of claim 9, wherein the first and second piezoelectric layer and all of the first and second IDTs are configured such that a respective radio frequency signal applied to each first and second IDT excites a respective primarily shear acoustic mode within the respective diaphragm.

11. The filter device of claim 9, wherein: the first acoustic resonators are shunt acoustic resonators of a ladder circuit, the second acoustic resonators are series acoustic resonators of the ladder circuit, and wherein the second thickness T**2** of the second piezoelectric layer is less than the first thickness T**1** of the first piezoelectric layer.

12. A method of fabricating an acoustic resonator device having a plurality of dies, the method comprising: forming one or more conductive vias through a first die, the first die having a plurality of shunt XBARs on a first surface, the plurality of shunt XBARs having a first piezoelectric layer with a first thickness T**1** and a first interdigital transducer having interleaved fingers on the first piezoelectric layer; and attaching a front surface of first die to a front surface of a second die, the second die having a plurality of series XBARs on a second surface, the plurality of series XBARs having a second piezoelectric layer with a second thickness T**2** that is less than the first thickness T**1** and a second interdigital transducer having interleaved fingers on the second piezoelectric layer, wherein the one or more conductive vias electrically connect the first XBARs to the second XBARs, and wherein the interleaved fingers of the first interdigital transducer have a different thickness than the interleaved fingers of the second interdigital transducer.

13. The method of claim 12, wherein the attaching of the front surface of first die includes: attaching bumps or contact pads of the die to bumps or contact pads of the second die; and ultrasonic bonding of the first die to the second die.

14. The method of claim 12, further comprising: forming the shunt XBARs on the first surface by forming the first interdigital transducer with the first interleaved fingers on first diaphragms formed over a first cavities of a first substrate; and forming the series XBARs on the second surface by forming the second interdigital transducer with the second interleaved fingers on second diaphragms formed over a second cavities of a second substrate.

15. The method of claim 12, wherein forming the one or more conductive vias includes: forming

wafer bumps or contact plating extending between and electrically connecting the plurality of shunt XBARs to the plurality of series XBARS, forming first through wafer vias (TWVs) for electrically connecting at least some of the plurality of series XBARs to external circuitry, and forming second TWVs for electrically connecting the plurality of shunt XBARS to ground.

16. The method of claim 15, wherein forming the wafer bumps or contact plating extending between and electrically connecting the plurality of shunt XBARs to the plurality of series XBARs includes: forming conductive vias that electrically connect the plurality of shunt XBARs to the plurality of series XBARs.

17. The method of claim 12, further comprising: forming one or more second conductive vias through the second die; and attaching a front surface of a third die to a back surface of the second die, the third die having a plurality of XBARs and a third surface, wherein the one or more second conductive vias electrically connect the plurality of shunt XBARs, the plurality of series XBARs and the plurality of XBARs of the third die.

18. The filter device of claim 1, wherein the second piezoelectric layer is formed of a different material than the first piezoelectric layer.

19. The filter device of claim 9, wherein the second piezoelectric layer is formed of a different material than the first piezoelectric layer.